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Module 7 Introduction

Lesson 1: Maximum Likelihood Inference

Lesson 2: Bayesian Inference

Module 7 Summative Assessment

Graded Assignment: Module 7 Summative Assessment

Submitted

Module 7 Summary

Your grade: 93.33%

Your score: 93.33% / Your highest score: 100%

Module 7 Summative Assessment

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Review Learning Objectives

Instructions

1.

Which of the following best describes the relationship between Bayesian inference and Bootstrapping?

Bayesian inference and bootstrapping have no connection and are used for entirely different purposes.

Bayesian inference and bootstrapping are both techniques for parameter estimation.

Bayesian inference is a technique for estimating the uncertainty of model parameters, while bootstrapping is a technique for quantifying the uncertainty of an estimator.

Bootstrapping is a technique used to generate more data, while Bayesian inference is a technique for dealing with pre-existing data.

Correct

Correct! Both Bayesian inference and bootstrapping are techniques for quantifying uncertainty, but they approach the problem from different angles.

2.

Consider the Bernoulli distribution, which has a single parameter p . How would you set up the likelihood function to estimate p using n observations?

$L(p) = \sum_{i=1}^n \log(p)^{y_i} (1-p)^{1-y_i}$

$L(p) = \sum_{i=1}^n \log(p) + (1-p)^{y_i}$

$L(p) = p^{y_i} / (1-p)^{y_i}$

$L(p) = p^{y_i} (1-p)^{1-y_i}$

Correct

Correct! This is the log likelihood function, not the likelihood function itself.

3.

How does Bayesian inference differ from frequentist inference in terms of handling uncertainty?

Bayesian inference does not handle uncertainty at all.

Frequentist inference quantifies uncertainty as a probability distribution over parameters while Bayesian inference uses confidence intervals.

Both Bayesian and frequentist inference handle uncertainty in the same way.

Bayesian inference quantifies uncertainty as a probability distribution over parameters while frequentist inference uses confidence intervals.

Correct

Correct! Bayesian inference quantifies uncertainty as a probability distribution over parameters.

4.

Which of the following is a limitation of maximum likelihood estimation?

It does not provide a measure of uncertainty for the parameter estimates.

All of the above.

It requires a large amount of computational resources.

It cannot handle missing data.

Correct

Correct! One limitation of maximum likelihood estimation is that it provides point estimates of the parameters, but does not provide a measure of uncertainty for these estimates.

5.

Which of the following Python code snippets correctly represents the implementation of a basic Bayesian inference procedure?

prior = [1/3, 1/3, 1/3] likelihood = [0.3, 0.4, 0.1] posterior = [p**1 / sum(p**1 for p in zip(prior, likelihood)) for p, l in zip(prior, likelihood)]

prior = np.array([1/3, 1/3, 1/3]) likelihood = np.array([0.3, 0.4, 0.1]) posterior = prior * likelihood / np.sum(prior * likelihood)

prior = np.array([1/3, 1/3, 1/3]) likelihood = np.array([0.3, 0.4, 0.1]) posterior = prior * likelihood

post = [1/3, 1/3, 1/3] likelihood = [0.3, 0.4, 0.1] posterior = [p**1 for p, l in zip(prior, likelihood)]

Correct

Correct! You have correctly applied the Bayesian inference procedure using numpy arrays and calculated the posterior by multiplying the prior and likelihood, then dividing by the sum.

6.

You are given a distribution of prior beliefs. After observing new data, how would you use Bayesian inference to update these beliefs?

Divide the prior by the likelihood of the data to get the posterior.

Multiply the prior by the likelihood of the data to get the posterior.

Subtract the likelihood of the data from the prior to get the posterior.

Add the prior to the likelihood of the data to get the posterior.

Correct

Correct! The posterior is obtained by multiplying the prior with the likelihood.

7.

In the context of maximum likelihood inference, what does the 'likelihood' refer to?

The probability of the model parameters given the data.

The posterior probability of the model parameters.

The probability of the data given the model parameters.

The accuracy of the model predictions.

Correct

Correct! The likelihood is the probability of observing the data given the model parameters.

8.

If you have a normal distribution and you want to estimate the mean and variance using maximum likelihood inference, which of the following is the correct likelihood function?

The sum of the absolute differences between each data point and the mean.

The sum of the squared differences between each data point and the mean.

The ratio of the sum of the data points to the number of data points.

The product of the probabilities of each data point given the mean and variance.

Correct

Correct! The likelihood function for a normal distribution is the product of the probabilities of each data point given the mean and variance.

9.

What does the 'maximum' in 'maximum likelihood estimation' refer to?

The maximum value of the likelihood function.

The maximum value of the parameters.

The maximum value of the data.

The maximum value of the prior distribution.

Correct

Correct! The 'maximum' in 'maximum likelihood estimation' refers to the maximum value of the likelihood function.

10.

Suppose you are using maximum likelihood inference to estimate the parameter of a Poisson distribution. Which of the following is the correct form of the likelihood function?

$L(\lambda) = \lambda^N \times e^{-\lambda}$

$L(\lambda) = \lambda^N \times \lambda!$

$L(\lambda) = \lambda^N \times e^{-\lambda} \times \lambda!$

$L(\lambda) = \lambda^N \times e^{-\lambda} \times \lambda!$

Correct

Correct! The likelihood function for the Poisson distribution is $(\lambda^N \times e^{-\lambda}) / N!$.

11.

In maximum likelihood inference, what is the consequence of choosing a model that is too complex for the data?

The model will overfit the data.

The likelihood will be maximized.

The model will underfit the data.

The likelihood will be minimized.

Correct

Correct! A model that is too complex for the data is likely to overfit, fitting the noise in the data as well as the underlying pattern.

12.

Which of the following situations is NOT suitable for using maximum likelihood estimation?

Estimating the mean of a sample with many outliers.

Estimating the parameter of an exponential distribution with a large sample size.

Estimating the mean of a sample drawn from a normal distribution.

Estimating the probability of success in a Bernoulli trial.

Correct

Correct! Maximum likelihood estimation is sensitive to outliers, so it might not be the best approach in this case.

13.

What is the primary goal of maximum likelihood inference?

To minimize the variance of parameter estimates.

To maximize the posterior probability of the parameters given the data.

To predict future observations.

To find the model parameters that maximize the likelihood of the observed data given the model.

Correct

Correct! In maximum likelihood inference, we aim to find the parameters that maximize the likelihood of the observed data given the model.

14.

Why is the Bayesian approach sometimes criticized for its subjectivity?

Because the likelihood function is chosen subjectively.

Because the prior distribution is derived from subjective data.

Because the prior distribution is chosen based on the researcher's subjective beliefs.

Because the Bayesian approach requires subjective coding skills.

Correct

Correct! The choice of the prior distribution in Bayesian inference is often based on the researcher's subjective beliefs.

15.

Which of the following steps would be necessary to create a Bayesian model?

Deriving a likelihood function and a prior distribution.

Determining the correlation coefficient.

Calculating the maximum likelihood estimate.

Performing a t-test.

Correct

Correct! Deriving a likelihood function and a prior distribution are essential steps in creating a Bayesian model.